coordinators and routers in a beacon mode enter into sleep state.

Here, coordinators are responsible for initiating and managing devices over the network. They wake up after regular intervals and transmit beacons to the routers in the network. Thus, they operate when communication needs results in lower duty cycles and longer battery usage.

In non-beacon mode, coordinators and routers continuously monitor the active state of incoming data and, hence, consume more power.

Zigbee can be used for a range of applications including remote patient monitoring, wireless lighting and electrical meters, traffic management systems, industrial automation, home automation and smart grid monitoring. Smart grid monitoring involves remote temperature monitoring, fault locating, reactive power management and so on.

It also helps retailers operate more efficiently by supporting just-in-time inventory practices, as well as monitoring temperatures, humidity, spills and so on.

The latest Zigbee 3.0, built on the existing Zigbee standard, unifies the market-specific application to allow all devices to be wirelessly connected in the same network.

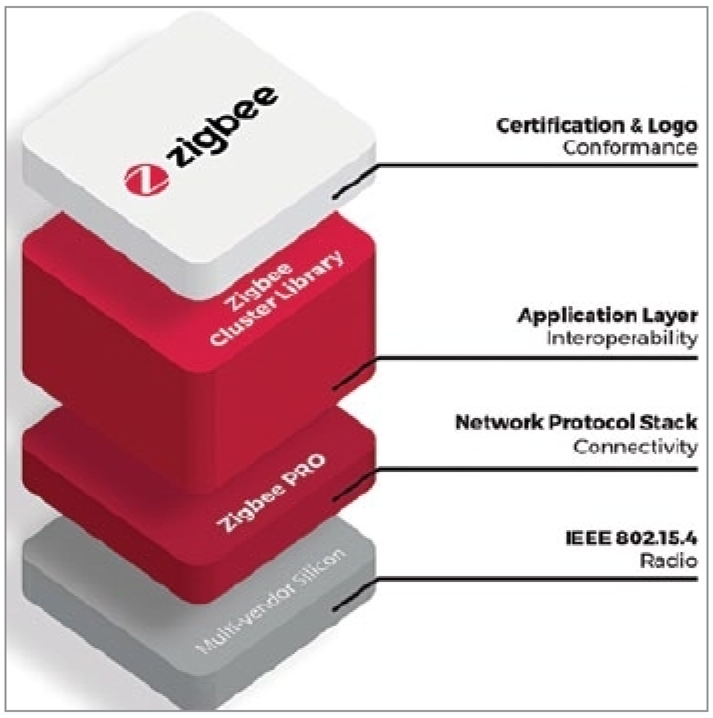
A Zigbee 3.0 certification scheme ensures interoperability of products from different manufacturers and 128-bit AES encryption for secure data connections.

RF transmission and SDR

RF is a form of electromagnetic transmission, ranging from 3kHz to 300GHz. It is used in wireless communication due to its property to penetrate through objects and travel long distances. Communication range depends on wavelength, transmitter power, receiver quality, and type, size and height of the antenna.

RF uses one transmitter section followed by a transmitting antenna, which is responsible for wireless signal transmission. This transmitted signal is received by the receiving antenna, amplified and demodulated to the original signal. This technology is widely used in military communications, personal communications, radio broadcast and communications in workplaces.

Software-defined radio (SDR) is a radio communication system that utilises software to implement components that have been implemented in the hardware transmitter/receiver unit on a computer or embedded system. SDR uses software to perform many of the basic functions on signals like mixing, amplification, filtering, modulation/demodulation, detection, etc.



Zigbee 3.0 (Credit: https://zigbee.org)

The software is easy to reconfigure and can be used on many platforms. Using the power of digital processing, SDR is being used for different applications in many different areas. The software is used to change the configuration of the radio or upgrade as per the function required at a given time.

It is possible to apply changes to any standard and even add new waveforms purely by upgrading the software and without the need for changes to the hardware. This can even be done remotely, thereby providing considerable savings in cost. It is also possible to reconfigure radios and change their performance by updating software.

Satellite communication

Microwaves are a form of electromagnetic transmission used in wireless communication systems. Wavelength of microwaves ranges from one metre to one millimetre. Frequency varies from 300MHz to 300GHz. These are widely used for long-distance communication and are relatively less expensive. Devices using satellite technology allow users to stay connected wirelessly from anywhere on Earth.

Satellite communication consists of space and ground segments. The signal sent to the satellite through a device (satellite phone) is amplified by the satellite and sent back to the receiver antenna, which is located on Earth.

Efforts are on to leverage the benefits of technology for the betterment of mankind. Important initiatives pursued by Indian Space Research Organisation towards societal development include tele-education, tele-medicine, village resource centre and disaster management system programmes. The potential of space technology for applications of national development and television broadcast is enormous.

Fixed satellite services support international Internet connectivity to private business networks. Mobile satellite services that use a variety of transportable receiver and transmitter equipment provide communication services for land mobile, maritime and aeronautical customers. Broadcast satellite services (BSS) offer high transmission power for reception using very small ground equipment.

LoRaWAN

Long Range (LoRa) technology is designed to wirelessly connect battery-operated things to the Internet in regional, national or global networks. It has large potential in smart cities, smart homes and buildings, smart agriculture, smart metering, smart supply chain and logistics, and many other applications.

LoRa connects devices up to fifteen kilometres apart in rural areas, and penetrates dense urban or deep indoor environments. This requires minimum energy, with a prolonged battery lifetime of up to ten years, minimising battery replacement costs.

LoRa features end-to-end AES128 encryption, mutual authentication, integrity protection and confidentiality. LoRa modulation is based on Chirp spread-spectrum technology, which makes it perform well with multipath fading, channel noise and Doppler effect, even at low power. LoRaWAN for wide-area networks can use channels with a bandwidth of 125kHz, 250kHz or 500kHz, depending on region or frequency plan. EFY